

Appendix A

Lagrangian and Hamiltonian equations of motion

Knowledge of Newton's equations of motion is sufficient to understand the basis of the Molecular Dynamics method. However, many of the more advanced simulation techniques make use of the Lagrangian or Hamiltonian formulations of classical mechanics. Here, we briefly sketch the relation between these different approaches (see also [712]). For a more detailed and more rigorous description of classical mechanics, the reader is referred to the book by Goldstein [54].

A.1 Action

The Lagrangian formulation of classical mechanics is based on a variational principle. The actual trajectory followed by a classical system in a time interval $\{t_b, t_e\}$, between an initial position x_b and a final position x_e , is the one for which the action, S , is an extremum (usually, a minimum). The classical action S for an arbitrary trajectory is defined as the time integral of the difference between the kinetic energy $\mathcal{K}$ and the potential energy $\mathcal{U}$ of the system, computed along that trajectory:

$$S = \int_{t_b}^{t_e} dt [\mathcal{K} - \mathcal{U}]. \quad (\text{A.1.1})$$

Before considering the general Lagrangian equations of motion that follow from this extremum principle, let us first consider a few simple examples.

The first case is that of a single particle that moves in the absence of an external potential, i.e., $\mathcal{U} = 0$. As the particle has to move from x_b to x_e in a time interval $t_e - t_b$, we already know its average velocity: v_{av} . If the particle would always move with this average velocity, it would follow a straight trajectory that we denote by $\bar{x}(t)$. Let us denote the true trajectory of the particle by $x(t) = \bar{x}(t) + \eta(t)$, where $\eta(t)$ is, as yet, unknown. Then the velocity of the particle is the sum of the average velocity v_{av} and the deviation from it, $\dot{\eta}(t)$:

$$v(t) = v_{av} + \dot{\eta}(t).$$

By construction,

$$\int dt \dot{\eta}(t) = 0.$$

In the present example, the potential energy is always zero, and hence the action S is determined by the time integral of the kinetic energy:

$$S = \frac{1}{2}m \int dt [v_{av} + \dot{\eta}(t)]^2 = S_{av} + \frac{1}{2}m \int dt \dot{\eta}^2(t).$$

Since the last term can never be less than zero, the action has its minimum if $\dot{\eta}(t) = 0$ for all t . In other words, we recover the well-known result that, in the absence of external forces, the particle moves with constant velocity. This is Newton's first law.

Next, consider a particle moving in a one-dimensional potential $U(x)$. In that case, the action is

$$S = \int_{t_b}^{t_e} dt \left[\frac{1}{2}m \left(\frac{dx(t)}{dt} \right)^2 - U(x) \right].$$

An arbitrary path, $x(t)$, can be written as the sum of the actual path that a classical particle will follow, $\bar{x}(t)$, plus the deviation from this path $\eta(t)$:

$$x(t) = \bar{x}(t) + \eta(t).$$

As before, we impose the initial and final positions of the particle and hence $\eta(t_b) = \eta(t_e) = 0$. For paths $x(t)$ that are close to the actual path, we can expand the action in powers of the (small) quantity $\eta(t)$. Actually, as $\eta(t)$ is itself a function of t , such an expansion is called a *functional expansion*. The action is extremal if the leading (linear) terms in this functional expansion vanish. Let us now consider the functional expansion of the action around the action of the true path to linear order in $\eta(t)$:

$$\begin{aligned} S &= \int_{t_b}^{t_e} dt \frac{1}{2}m \left(\frac{d\bar{x}(t)}{dt} + \frac{d\eta(t)}{dt} \right)^2 - U[\bar{x}(t) + \eta(t)] \\ &\approx \int_{t_b}^{t_e} dt \frac{1}{2}m \left[\left(\frac{d\bar{x}(t)}{dt} \right)^2 + 2 \frac{d\bar{x}(t)}{dt} \frac{d\eta(t)}{dt} \right] - \left[U(\bar{x}(t)) + \frac{\partial U(\bar{x})}{\partial x} \eta(t) \right] \\ &= \bar{S} + \int_{t_b}^{t_e} dt \left[m \frac{d\bar{x}(t)}{dt} \frac{d\eta(t)}{dt} - \frac{\partial U(\bar{x})}{\partial x} \eta(t) \right] \\ &= \bar{S} + m \frac{d\bar{x}(t)}{dt} \eta(t) \Big|_{t_b}^{t_e} - \int_{t_b}^{t_e} dt \left[m \frac{d^2 \bar{x}(t)}{dt^2} + \frac{\partial U(\bar{x})}{\partial x} \right] \eta(t), \end{aligned}$$

where the last step has been obtained via partial integration. Since by definition, $\eta(t) = 0$ at the boundaries, the second term on the right-hand side vanishes. The action has its extremum if the integrand in the last line of the above equation vanishes for arbitrary $\eta(t)$. This condition can be satisfied if and only if

$$m \frac{d^2 \bar{x}(t)}{dt^2} = - \frac{\partial U(\bar{x})}{\partial x}, \quad (\text{A.1.2})$$

which is Newton's second law. In other words, Newton's equations of motion can be derived from the statement that a particle follows a path for which the action is an extremum.

A.2 Lagrangian

There would be little point in introducing this alternative expression of the laws of classical mechanics if it did not allow us to do more than simply rederive $F = ma$. In fact, the Lagrangian formulation of classical mechanics turns out to be very powerful. For one thing, the Lagrangian approach makes it easy to derive equations of motion in non-Cartesian coordinate frames. Suppose that we wish to use some generalized coordinates q instead of the Cartesian coordinate x . For example, consider a pendulum of length l in a uniform gravitational field. The angle that the pendulum makes with the vertical (i.e., with the direction of the gravitational field) can be used to specify its orientation. Since the path that the pendulum follows is clearly independent of the coordinates that we happen to use to specify its state, the action, S , should be the same:

$$S = \int dt \mathcal{L}(x, \dot{x}) = \int dt \mathcal{L}(q, \dot{q}), \quad (\text{A.2.1})$$

where the quantity $\mathcal{L}$ is called the Lagrangian. The Lagrangian is defined as the kinetic energy minus the potential energy¹:

$$\mathcal{L} \equiv \mathcal{K}(\dot{q}) - \mathcal{U}(q). \quad (\text{A.2.2})$$

We again introduce our actual path $\bar{q}(t)$ and the deviation $\eta(t)$ from it:

$$\begin{aligned} q(t) &= \bar{q}(t) + \eta(t) \\ \dot{q}(t) &= \dot{\bar{q}}(t) + \dot{\eta}(t). \end{aligned}$$

We can write for the Lagrangian, $\mathcal{L}$,

$$\mathcal{L}(q, \dot{q}) = \mathcal{L}(\bar{q}, \dot{\bar{q}}) + \frac{\partial \mathcal{L}(\bar{q}, \dot{\bar{q}})}{\partial \dot{q}} \dot{\eta}(t) + \frac{\partial \mathcal{L}(\bar{q}, \dot{\bar{q}})}{\partial q} \eta(t).$$

As in the previous section, we use the functional expansion of S in powers of $\eta(t)$ to derive an expression for the classical path. To this end, we substitute the Lagrangian in the expression for the action (A.2.1). Next, we write a possible path of the particle as the sum of the actual path and a correction $\eta(t)$. As before, we use partial integration and use the fact that $\eta(t)$ vanishes at the boundaries of the integration. It then follows that the action has an extremum if

$$\int dt \left[-\frac{d}{dt} \left(\frac{\partial \mathcal{L}(\bar{q}, \dot{\bar{q}})}{\partial \dot{q}} \right) + \frac{\partial \mathcal{L}(\bar{q}, \dot{\bar{q}})}{\partial q} \right] \eta(t) = 0, \quad (\text{A.2.3})$$

¹ The correct definition is more restrictive; see [54] for more details.

which is satisfied for arbitrary $\eta(t)$ if and only if

$$\left[-\frac{d}{dt} \left(\frac{\partial \mathcal{L}(\bar{q}, \dot{\bar{q}})}{\partial \dot{\bar{q}}} \right) + \frac{\partial \mathcal{L}(\bar{q}, \dot{\bar{q}})}{\partial \bar{q}} \right] = 0. \quad (\text{A.2.4})$$

This is the Lagrangian equation of motion. To cast this equation of motion in a more familiar form, we introduce the generalized momentum p associated with the generalized coordinate q :

$$p \equiv \frac{\partial \mathcal{L}(q, \dot{q})}{\partial \dot{q}}. \quad (\text{A.2.5})$$

Substitution of this expression into Eq. (A.2.4) yields

$$\dot{p} = \frac{\partial \mathcal{L}(q, \dot{q})}{\partial q}. \quad (\text{A.2.6})$$

As the above formulation is valid for any coordinate system, it should certainly hold for Cartesian coordinates. In these coordinates the Lagrangian reads

$$\mathcal{L}(x, \dot{x}) = \frac{1}{2} m \dot{x}^2 - U(x).$$

The momentum associated with x is

$$p_x = \frac{\partial \mathcal{L}(x, \dot{x})}{\partial \dot{x}} = m \dot{x}$$

and the equation of motion is

$$m \ddot{x} = -\frac{\partial U(x)}{\partial x},$$

which is indeed the result we would obtain from Newton's equation of motion.

Illustration 24 (A pendulum in a gravitational field). Consider a simple pendulum of length l with mass m (see Fig. A.1). A uniform gravitational field is acting on the pendulum and the potential energy is a simple function of the angle θ that the pendulum makes with the vertical:

$$U(\theta) = mgl[1 - \cos(\theta)].$$

We wish to express the equations of motion in terms of the generalized coordinate θ . The Lagrangian, $\mathcal{L}$, is

$$\begin{aligned} \mathcal{L} &= \mathcal{K} - \mathcal{U} = \frac{1}{2} m [\dot{x}^2(t) + \dot{y}^2(t)] - U(\theta) \\ &= \frac{ml^2}{2} \dot{\theta}^2 - U(\theta). \end{aligned}$$

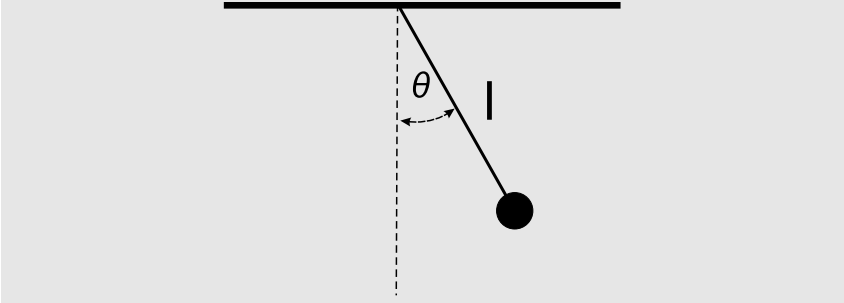


FIGURE A.1 A simple pendulum of length l with mass m .

The generalized momentum is defined as

$$p_\theta = \frac{\partial \mathcal{L}}{\partial \dot{q}} = ml^2 \dot{\theta}$$

and the equation of motion follows from Eq. (A.2.6)

$$\dot{p}_\theta = - \frac{\partial U(\theta)}{\partial \theta}$$

or

$$\ddot{\theta} = - \frac{1}{ml^2} \frac{\partial U(\theta)}{\partial \theta}.$$

A.3 Hamiltonian

Using the Lagrangian, we obtain equations of motion in terms of q and $\dot{q}$. Often, it is convenient to express the equations of motion in terms of q and its conjugate momentum p . To do this, we can perform a Legendre transformation²:

$$\mathcal{H}(q, p) \equiv p\dot{q} - \mathcal{L}(q, \dot{q}, t). \quad (\text{A.3.1})$$

² In thermodynamics, Legendre transforms are used to derive various thermodynamic potentials. For example, the energy E is a natural function of the entropy S and volume V : $E = E(S, V)$, i.e., in these variables, E is a thermodynamic potential. In most practical applications, it is more convenient to have the temperature T rather than the entropy S as the independent variable. Since the temperature is the variable conjugate to the entropy ($\partial E / \partial S = T$), we can perform a Legendre transform to remove the S dependence:

$$A \equiv E - TS,$$

yielding

$$dA = dE - d(TS) = -SdT - pdV.$$

For historical reasons, the Legendre transform linking the Lagrangian to the Hamiltonian has the opposite sign.

This equation defines the *Hamiltonian* $\mathcal{H}$ of the system. As $\mathcal{H}$ is a function of q , p , and, in general, also of t , it is clear that we can write an infinitesimal variation of $\mathcal{H}$ as

$$d\mathcal{H}(q, p) = \frac{\partial \mathcal{H}}{\partial p} dp + \frac{\partial \mathcal{H}}{\partial q} dq + \frac{\partial \mathcal{H}}{\partial t} dt. \quad (\text{A.3.2})$$

But, using the definition of $\mathcal{H}$, we can also write

$$\begin{aligned} d\mathcal{H}(q, p) &= d(p\dot{q}) - d\mathcal{L}(q, \dot{q}) \\ &= p d\dot{q} + \dot{q} dp - \left[\frac{\partial \mathcal{L}}{\partial q} dq + \frac{\partial \mathcal{L}}{\partial \dot{q}} d\dot{q} + \frac{\partial \mathcal{L}}{\partial t} dt \right] \\ &= p d\dot{q} + \dot{q} dp - \dot{p} dq - p d\dot{q} - \frac{\partial \mathcal{L}}{\partial t} dt \\ &= \dot{q} dp - \dot{p} dq - \frac{\partial \mathcal{L}}{\partial t} dt, \end{aligned}$$

where we have used the definitions of p and $\dot{p}$, Eqs. (A.2.5) and (A.2.6), respectively. It then follows directly that

$$\frac{\partial \mathcal{H}}{\partial p} = \dot{q} \quad (\text{A.3.3})$$

$$\frac{\partial \mathcal{H}}{\partial q} = -\dot{p}. \quad (\text{A.3.4})$$

These are the desired equations of motion in terms of q , p . For most systems that we consider in this book, the Lagrangian does not explicitly depend on time. Under those circumstances, the Hamiltonian is conserved. This follows directly from the equations of motion:

$$\begin{aligned} \frac{d\mathcal{H}(q, p)}{dt} &= \frac{\partial \mathcal{H}}{\partial p} \dot{p} + \frac{\partial \mathcal{H}}{\partial q} \dot{q} \\ &= -\frac{\partial \mathcal{H}}{\partial p} \frac{\partial \mathcal{H}}{\partial q} + \frac{\partial \mathcal{H}}{\partial q} \frac{\partial \mathcal{H}}{\partial p} \\ &= 0. \end{aligned}$$

This conservation law expresses the fact that, in a closed system, the total energy is conserved. In Cartesian coordinates, the Hamiltonian can be written as

$$\begin{aligned} \mathcal{H}(x, p_x) &= \dot{x} p_x - \mathcal{L}(x, \dot{x}) \\ &= m\dot{x}^2 - \frac{1}{2}m\dot{x}^2 + U(x) \\ &= \frac{1}{2m} p_x^2 + U(x), \end{aligned}$$

and the Hamiltonian equations of motion reduce to Newton's equations

$$\begin{aligned}\dot{x} &= \frac{\partial \mathcal{H}}{\partial p_x} = \frac{p_x}{m} \\ \dot{p}_x &= -\frac{\partial \mathcal{H}}{\partial x} = -\frac{\partial U(x)}{\partial x}.\end{aligned}$$

The Hamiltonian equations of motion are two first-order differential equations—one for p and one for q . In contrast, the Lagrangian formalism yields a single second-order equation. However, both formalisms yield identical results. The choice between the two is dictated by considerations of mathematical convenience.

Example 28 (A pendulum in a gravitational field: Part II). We consider again the simple pendulum in a uniform gravitational field, introduced in Example 24:

$$U(\theta) = mgl[1 - \cos(\theta)],$$

where θ is the angle that the pendulum makes with the vertical and g the gravitational acceleration.

In Example 24 we have derived the equations of motion from the Lagrangian in terms of a second-order differential equation in θ . Now we will use Hamilton's formulation.

The Lagrangian is

$$\mathcal{L}(\theta, \dot{\theta}) = \mathcal{U}_K - \mathcal{U}_P = \frac{ml^2}{2} \dot{\theta}^2 - U(\theta).$$

The Lagrangian depends on the variables θ and $\dot{\theta}$ and in the Hamiltonian language we want to express the equations of motion in terms of θ and its conjugate momentum p_θ . This conjugate momentum is defined by Eq. (A.2.5)

$$p_\theta \equiv \frac{\partial \mathcal{L}(\theta, \dot{\theta})}{\partial \dot{\theta}} = ml^2 \dot{\theta}.$$

The Hamiltonian follows from the Legendre transformation (A.3.1)

$$\begin{aligned}\mathcal{H} &= p_\theta \dot{\theta} - \mathcal{L}(\theta, \dot{\theta}) \\ &= \frac{p_\theta}{2ml^2} + \mathcal{U}(\theta) \\ &= \frac{1}{2} ml^2 \dot{\theta}^2 + U(\theta),\end{aligned}$$

which is, of course, equal to the total energy of the pendulum.

The equations of motion follow from Eqs. (A.3.3) and (A.3.4):

$$\dot{\theta} = \frac{\partial \mathcal{H}}{\partial p_\theta} = \frac{p_\theta}{ml^2}$$

$$\dot{p}_\theta = -\frac{\partial \mathcal{H}}{\partial \theta} = -\frac{dU(\theta)}{d\theta},$$

which are the desired equations of motion in terms of two first-order differential equations.

A.4 Hamilton dynamics and statistical mechanics

The choice between the Hamiltonian and Lagrangian formulations of classical mechanics is determined by considerations of convenience. One example of a case where the Lagrangian formalism is more convenient is in the derivation of the equations of motion of a system with constraints (see section 14.1). On the other hand, the Hamiltonian expressions are to be used when establishing the connection with statistical mechanics (see Chapter 2).

A.4.1 Canonical transformation

In the Hamiltonian formulation, the generalized coordinates and momenta are independent variables. One can therefore introduce a transformation of both variables simultaneously. For example, the transformation of the coordinates q, p to Q, P is denoted by

$$\begin{aligned} Q &= Q(q, p) \\ P &= P(q, p) \end{aligned} \tag{A.4.1}$$

and the inverse transformation, Q, P into q, p , by

$$\begin{aligned} q &= q(Q, P) \\ p &= p(Q, P). \end{aligned} \tag{A.4.2}$$

Obviously, the value of any function of the phase-space coordinates is unaffected by the coordinate transformation. In the case of the Hamiltonian, this implies that

$$\mathcal{H}(q, p) \equiv \mathcal{H}[Q(p, q), P(q, p)] \equiv \mathcal{H}'(Q, P). \tag{A.4.3}$$

In general, the equations of motion in the new coordinates are not of the canonical form, unless the coordinate transformation is *canonical*.³ If the coordinate transformation is canonical, the equations of motion for the new phase-space coordinates Q, P are

$$\dot{Q} = \left(\frac{\partial \mathcal{H}'(Q, P)}{\partial P} \right) \tag{A.4.4}$$

³ As we assume that time does not appear explicitly in these equations, we are defining a so-called restricted canonical transformation.

$$\dot{P} = - \left(\frac{\partial \mathcal{H}'(Q, P)}{\partial Q} \right). \quad (\text{A.4.5})$$

From Eq. (A.4.1) and the Hamilton equations of motion for the coordinates q , p , it follows that

$$\begin{aligned} \dot{Q} &= \left(\frac{\partial Q(q, p)}{\partial q} \right) \dot{q} + \left(\frac{\partial Q(q, p)}{\partial p} \right) \dot{p} \\ &= \left(\frac{\partial Q(q, p)}{\partial q} \right) \left(\frac{\partial \mathcal{H}(q, p)}{\partial p} \right) - \left(\frac{\partial Q(q, p)}{\partial p} \right) \left(\frac{\partial \mathcal{H}(q, p)}{\partial q} \right). \end{aligned}$$

Using Eq. (A.4.3), we can write

$$\left(\frac{\partial \mathcal{H}'(Q, P)}{\partial P} \right) = \left(\frac{\partial \mathcal{H}(q, p)}{\partial p} \right) \left(\frac{\partial p(P, Q)}{\partial P} \right) + \left(\frac{\partial \mathcal{H}(q, p)}{\partial q} \right) \left(\frac{\partial q(P, Q)}{\partial P} \right).$$

This equation can only be equal to expression (A.4.4) for $\dot{Q}$ if

$$\left(\frac{\partial Q(q, p)}{\partial q} \right) = \left(\frac{\partial p(Q, P)}{\partial P} \right) \quad \text{and} \quad \left(\frac{\partial Q(q, p)}{\partial p} \right) = - \left(\frac{\partial q(Q, P)}{\partial P} \right). \quad (\text{A.4.6})$$

Similarly, we can start with $\dot{P}$, and derive two other conditions:

$$\left(\frac{\partial P(q, p)}{\partial q} \right) = - \left(\frac{\partial p(Q, P)}{\partial Q} \right) \quad \text{and} \quad \left(\frac{\partial P(q, p)}{\partial p} \right) = \left(\frac{\partial q(Q, P)}{\partial Q} \right). \quad (\text{A.4.7})$$

These two equations define the condition for a canonical transformation.

A.4.2 Symplectic condition

We can express the above conditions for a canonical transformation in a single equation, by using a matrix notation. Let $\mathbf{\Gamma}$ be a $2dN$ -dimensional vector containing the generalized coordinates q_i and momenta p_i of the N particles in d dimensions (see Section 2.5.1). Hamilton's equations of motion (A.3.3) and (A.3.4) can be written as

$$\dot{\mathbf{\Gamma}} = \boldsymbol{\omega} \frac{\partial \mathcal{H}}{\partial \mathbf{\Gamma}}, \quad (\text{A.4.8})$$

where $\boldsymbol{\omega}$ is an antisymmetric matrix defined as

$$\boldsymbol{\omega} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}.$$

In a similar way, we can define $\boldsymbol{\xi}$ to be the $2N$ -dimensional vector containing the generalized coordinates Q_i and P_i . Using the matrix notation the transfor-

mation (A.4.1) from Q, P to q, p is written as

$$\xi = \xi(\Gamma).$$

For the time derivatives of ξ , we can write

$$\dot{\xi} = M\dot{\Gamma},$$

where M is the Jacobian matrix of the transformation. The elements of this matrix are

$$M_{ij} = \frac{\partial \xi_i}{\partial \Gamma_j}. \quad (\text{A.4.9})$$

We can write, using Eq. (A.4.8), for the time derivatives of ξ

$$\dot{\xi} = M\omega \frac{\partial \mathcal{H}}{\partial \Gamma}. \quad (\text{A.4.10})$$

In a similar way, we can define the inverse transformation (A.4.2)

$$\Gamma = \Gamma(\xi).$$

Since

$$\mathcal{H}(p, q) = \mathcal{H}(P, Q),$$

we can write

$$\frac{\partial \mathcal{H}(\Gamma)}{\partial \Gamma_i} = \sum_j \frac{\partial \mathcal{H}(\xi)}{\partial \xi_j} \frac{\partial \xi_j}{\partial \Gamma_i}. \quad (\text{A.4.11})$$

If we define the transposed matrix⁴ of M as defined in Eq. (A.4.9),

$$\tilde{M}_{ij} = \frac{\partial \xi_j}{\partial \Gamma_i}.$$

This allows us to rewrite Eq. (A.4.11) in matrix notation as

$$\frac{\partial \mathcal{H}(\Gamma)}{\partial \Gamma} = \tilde{M} \frac{\partial \mathcal{H}(\xi)}{\partial \xi}. \quad (\text{A.4.12})$$

If we combine Eqs. (A.4.10) and (A.4.12), we have

$$\dot{\xi} = M\omega \tilde{M} \frac{\partial \mathcal{H}}{\partial \xi}.$$

This expression for the equations of motion is valid for any set of variables ξ that are being transformed (independently of time) from the set Γ . Such a

⁴ One can obtain the transposed matrix of a given matrix A by interchanging rows and columns, i.e., $\tilde{a}_{ij} = a_{ji}$.

transformation is canonical if the equations of motion in the new coordinates have the canonical form:

$$\dot{\xi} = \omega \frac{\partial \mathcal{H}}{\partial \xi}.$$

This can only be the case if M satisfies the condition

$$M\omega\tilde{M} = \omega. \quad (\text{A.4.13})$$

This condition is often called the *symplectic condition*. A matrix M that satisfies this condition is called a *symplectic matrix*.⁵

A.4.3 Statistical mechanics

Using the symplectic notation for a canonical transformation, we consider the implications for statistical mechanics. In the microcanonical ensemble, the classical partition function Ω of a three-dimensional atomic system is defined as

$$\Omega_{N,V,E} = \frac{1}{h^{3N}N!} \int d\mathbf{p}^N d\mathbf{q}^N \delta(\mathcal{H}(p, q) - E), \quad (\text{A.4.14})$$

where h is Planck's constant and the delta-function restricts the integration to the hypersurface in phase space defined by $\mathcal{H}(p, q) = E$. We can re-express this integral in terms of other phase-space coordinates, but then we have to take into account that a volume element in the two coordinate sets needs not to be the same. The volume element associated with Γ is

$$d\Gamma = dq_1 \dots dq_N dp_1 \dots dp_N$$

and to ξ

$$d\xi = dQ_1 \dots dQ_N dP_1 \dots dP_N.$$

These two volume elements are related via the Jacobian matrix of the transformation matrix

$$d\Gamma = |\text{Det}(M)| d\xi. \quad (\text{A.4.15})$$

This equation shows that, in general, a coordinate transformation will result in the appearance of a Jacobian in the partition function:

$$\Omega_{N,V,E} = \frac{1}{h^{3N}N!} \int d\mathbf{P}^N d\mathbf{Q}^N |\text{Det}(M)| \delta(\mathcal{H}'(P, Q) - E). \quad (\text{A.4.16})$$

⁵ To see that this condition is identical to Eqs. (A.4.6) and (A.4.7), we have to multiply this equation from the right with the inverse matrix of M :

$$M\omega = \omega\tilde{M}^{-1}.$$

When computing ensemble averages in coordinate systems other than the original Cartesian one, the Jacobian of the transformation, M , may be different from one, and should be taken into account. In what follows, we denote the Jacobian $|\text{Det}(M)|$ by the symbol ω .

For a canonical transformation, i.e., obeys condition (A.4.13), the absolute value of the Jacobian is one. To derive this result, we take the determinant of both sides of the symplectic condition (A.4.13)

$$\begin{aligned}\text{Det}(M\omega\tilde{M}) &= \text{Det}(\omega) \\ \text{Det}^2(M)\text{Det}(\omega) &= \text{Det}(\omega).\end{aligned}$$

This equation can only be true if the determinant of M is ± 1 , which implies that for a canonical transformation the absolute value of the Jacobian associated with this transformation must be one.

The natural time evolution in phase space of a classical system may be considered as a coordinate transformation:

$$\Gamma(t_0) \rightarrow \Gamma(t).$$

One important property of a Hamiltonian system is that the natural time evolution corresponds to a symplectic coordinate transformation. We can consider the transformation from $\Gamma(t_0)$ to $\Gamma(t)$ as a sequence of infinitesimal transformations with time step δt . Suppose that we define the evolution of the coordinates during the time interval δt as a transformation of coordinates from Γ to ξ :

$$\begin{aligned}\xi &= \xi(\Gamma) \\ &= \Gamma(t + \delta t) \\ &= \Gamma(t) + \dot{\Gamma}(t)\delta t.\end{aligned}$$

The Jacobian of this transformation is

$$\begin{aligned}M &\equiv \frac{\partial \xi}{\partial \Gamma} \\ &= 1 + \delta t \frac{\partial}{\partial \Gamma} \left(\omega \frac{\partial \mathcal{H}}{\partial \Gamma} \right) \\ &= 1 + \delta t \omega \frac{\partial^2 \mathcal{H}}{\partial \Gamma \partial \Gamma},\end{aligned}$$

where

$$\left(\frac{\partial^2 \mathcal{H}}{\partial \Gamma \partial \Gamma} \right)_{ij} = \frac{\partial^2 \mathcal{H}}{\partial \Gamma_i \partial \Gamma_j}.$$

Taking into account that ω is an antisymmetric matrix, we can write for the transpose of the matrix M :

$$\tilde{M} = 1 - \frac{\partial^2 \mathcal{H}}{\partial \mathbf{\Gamma} \partial \mathbf{\Gamma}} \omega.$$

Substitution of this expression for the Jacobian into the symplectic condition (A.4.13) yields (to first order in δt)

$$\begin{aligned} M\omega\tilde{M} &= \left(1 + \delta t \omega \frac{\partial^2 \mathcal{H}}{\partial \mathbf{\Gamma} \partial \mathbf{\Gamma}}\right) \omega \left(1 - \delta t \frac{\partial^2 \mathcal{H}}{\partial \mathbf{\Gamma} \partial \mathbf{\Gamma}} \omega\right) \\ &\approx \omega + \delta t \omega \frac{\partial^2 \mathcal{H}}{\partial \mathbf{\Gamma} \partial \mathbf{\Gamma}} \omega - \omega \delta t \frac{\partial^2 \mathcal{H}}{\partial \mathbf{\Gamma} \partial \mathbf{\Gamma}} \omega \\ &= \omega. \end{aligned}$$

Hence the symplectic condition holds for the evolution of $\mathbf{\Gamma}$ during an infinitesimal time interval. As we can consider the evolution of $\mathbf{\Gamma}$ during a finite interval, as a sequence of canonical transformations of infinitesimal steps, the total time evolution also satisfies the symplectic condition.

One may view the Hamiltonian as the generator of a canonical transformation acting on all points in phase space. As the Jacobian of a canonical transformation is equal to 1, the size of a volume element in phase space does not change during the natural time evolution of a Hamiltonian system. Moreover, the density $f(q(t), p(t))$ around any point in phase space also remains constant during the time evolution. To see this, consider a volume V in phase space bounded by a surface S . During time evolution, the surface moves, and so do all points inside the surface. However, a point cannot *cross* the surface. The reason is simple: if two trajectories in phase space crossed, it would imply that two trajectories start from the same phase-space point. But this is impossible, as it would mean that a trajectory starting from this point is not uniquely specified by its initial conditions. Hence, the number of phase-space points inside any volume does not change in time. As the volume itself is also constant, this implies that the phase-space density (i.e., the number of points per unit volume) is constant. In other words: the phase-space density of a Hamiltonian system behaves like an incompressible fluid:

$$\frac{df}{dt} = 0. \quad (\text{A.4.17})$$

While the exact solution of Hamilton's equations of motion will satisfy the incompressibility condition, discrete, numerical schemes will—in general—violate it. As before, we can consider any numerical MD algorithm (e.g., Verlet, velocity Verlet, ...) as a transformation from $(q(t), p(t))$ to $(q(t + \Delta t), p(t + \Delta t))$. We can then compute the Jacobian of this transformation, and

check whether it is equal to 1 (see sections 4.3 and 4.3.4). For all “good” algorithms to solve Newton’s equations of motion, the Jacobian of the transformation from $(q(t), p(t))$ to $(q(t + \Delta t), p(t + \Delta t))$ is equal to 1 —such algorithms are said to be “area-preserving.” It should be noted that the symplectic condition implies more than just the area-preserving properties. Unfortunately, these other consequences do not have such a simple intuitive interpretation. When we say that it is desirable that an algorithm be symplectic, we mean more than that it should be area-preserving —it should really satisfy the symplectic condition. Fortunately, in many cases, the symplectic nature of an algorithm is easy to demonstrate by making use of the fact that *any* set of classical Hamiltonian equations of motion satisfies the symplectic condition. An algorithm that can be written as a sequence of exact time evolutions generated by simple Hamiltonians is, therefore, necessarily symplectic. An example is the Verlet algorithm. As discussed in section 4.3.4, this algorithm can be viewed as a sequence of exact propagations using either the kinetic part of the Hamiltonian or the potential part. Either propagation satisfies the symplectic condition. Hence the Verlet algorithm as a whole is symplectic. For an accessible discussion of symplectic dynamics in general, see ref. [713]. A discussion of symplectic integrators for Molecular Dynamics simulations can be found in ref. [714].